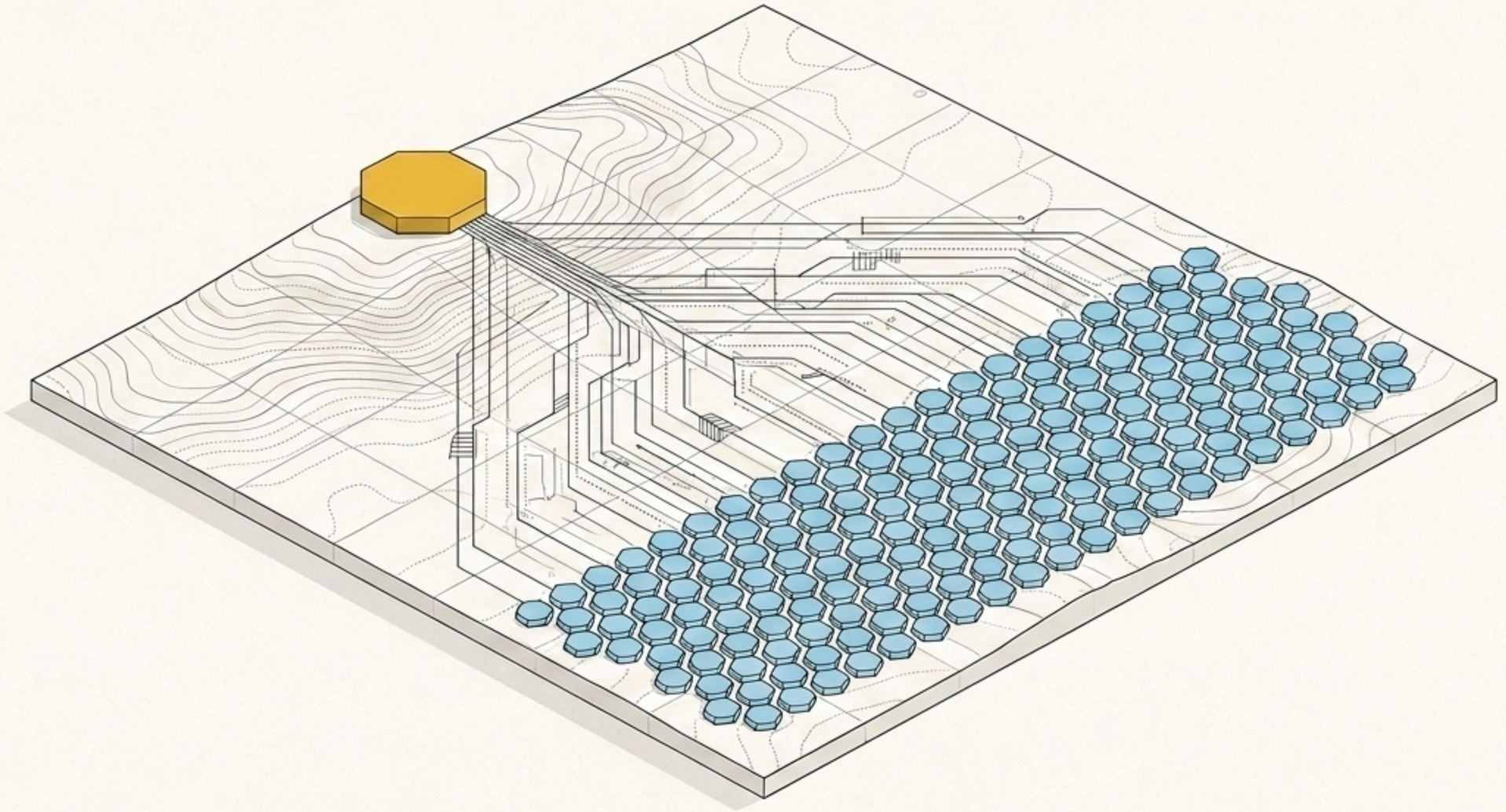
SCALING GROUND TRUTH

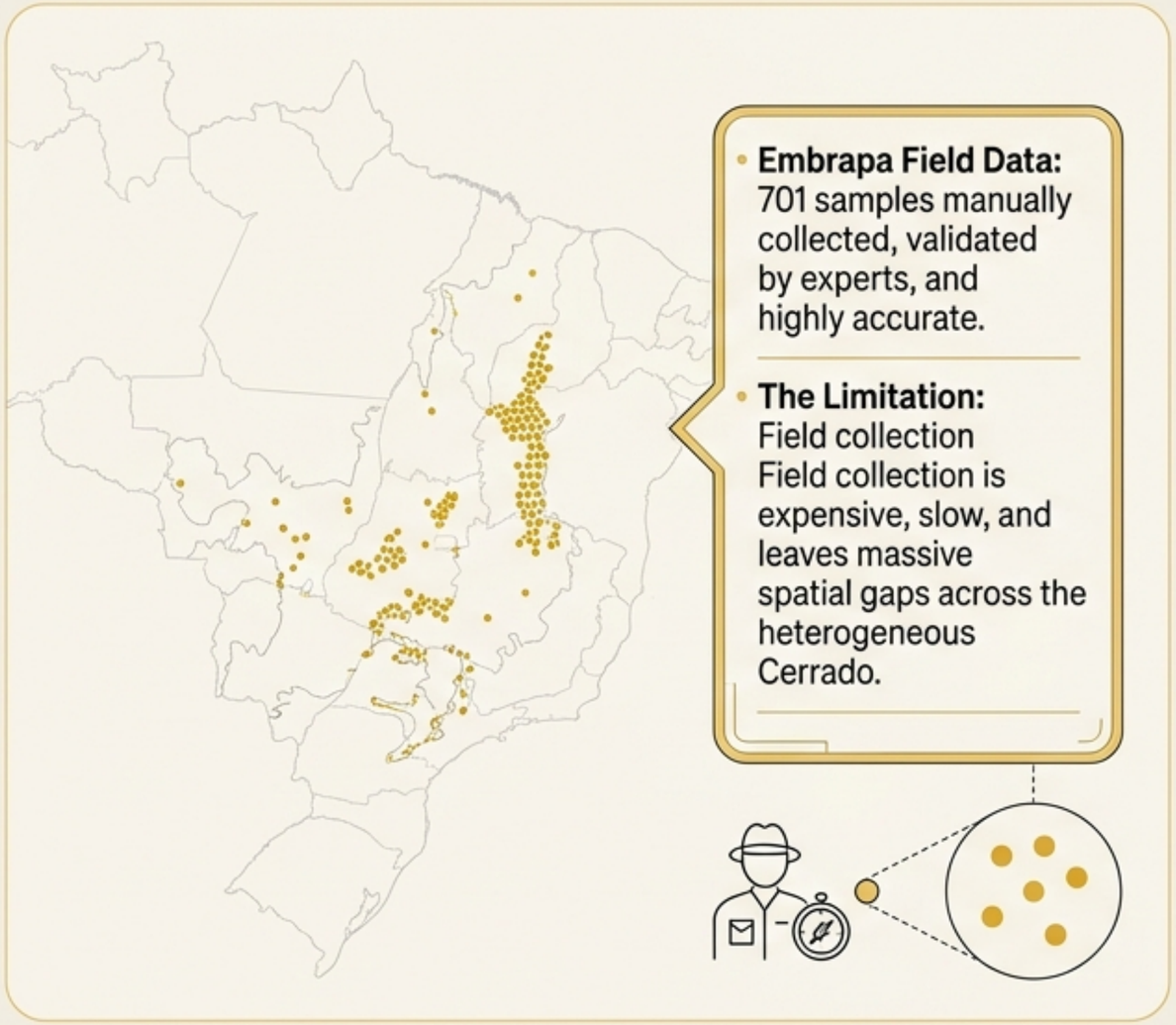
A Dual-Filter Methodology for Augmenting Pasture Classification Samples

LAPIG / UFG — WRI Brasil



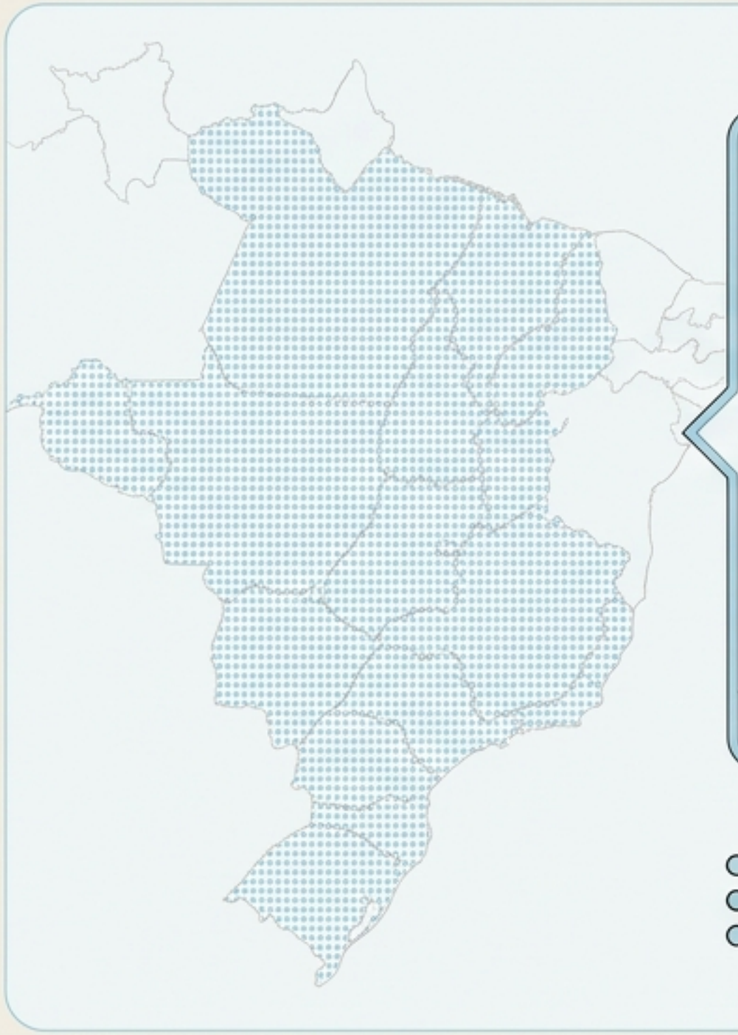
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THE GOLD STANDARD

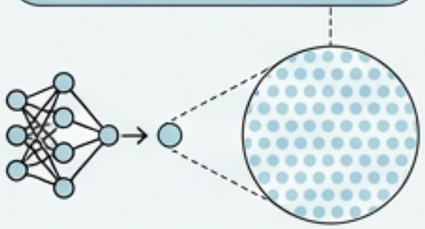


- **Embrapa Field Data:** 701 samples manually collected, validated by experts, and highly accurate.
- **The Limitation:** Field collection Field collection is expensive, slow, and leaves massive spatial gaps across the heterogeneous Cerrado.

THE ML REQUIREMENT



- **MapBiomas Candidates:** 62,009 unverified sample points distributed across the biome.
- **The Goal:** Transfer the reliability of the 701 ground truth points to a massive 50,000+ point dataset required to train the 10m Random Forest generator.

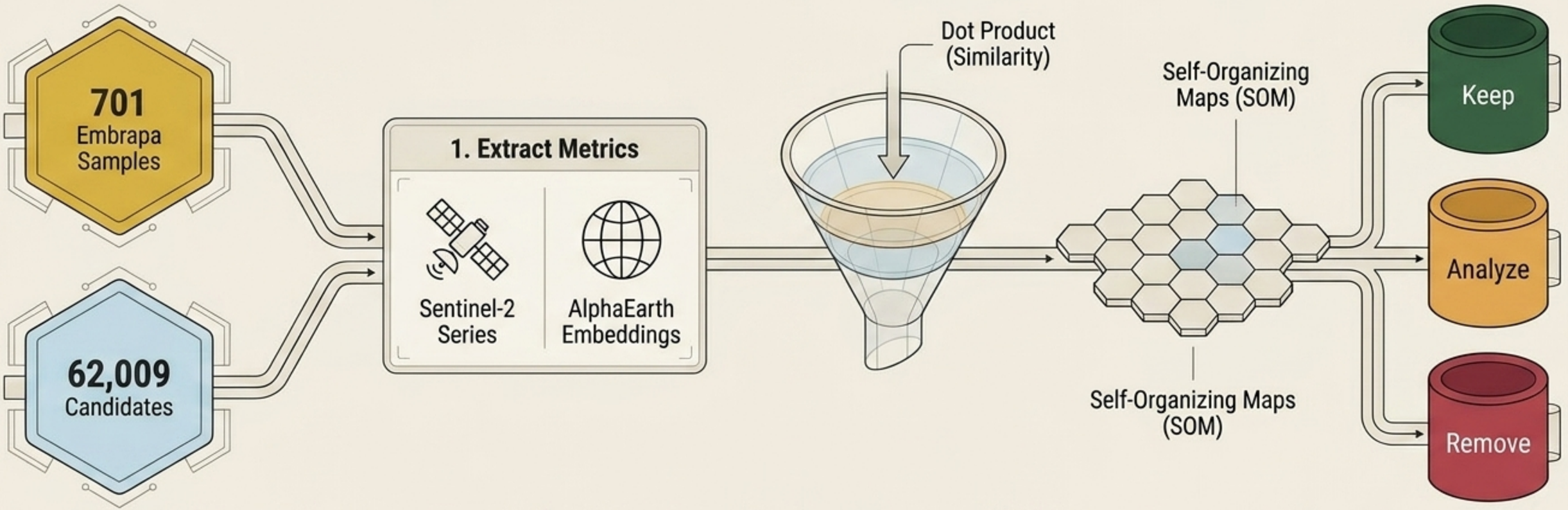


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INPUT PHASE

2. Phase 1 (Spatial Filter)

3. Phase 2 (Temporal Filter)

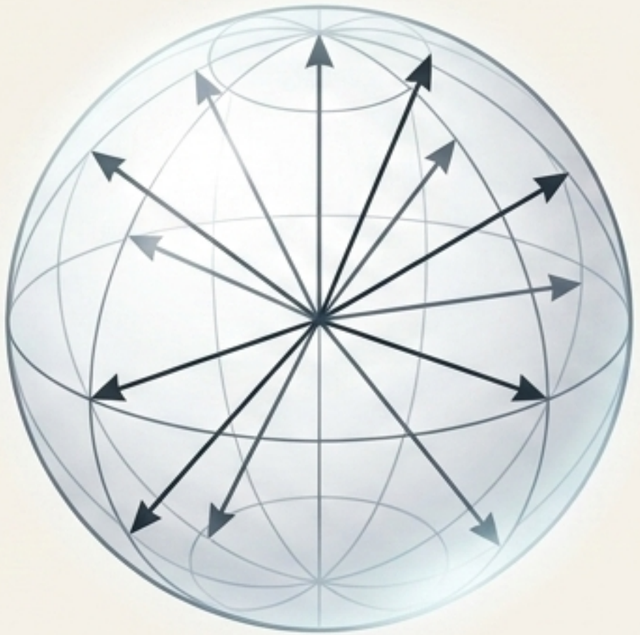


Extract spatial and temporal signatures → Filter by structural similarity → Validate by phenological coherence.

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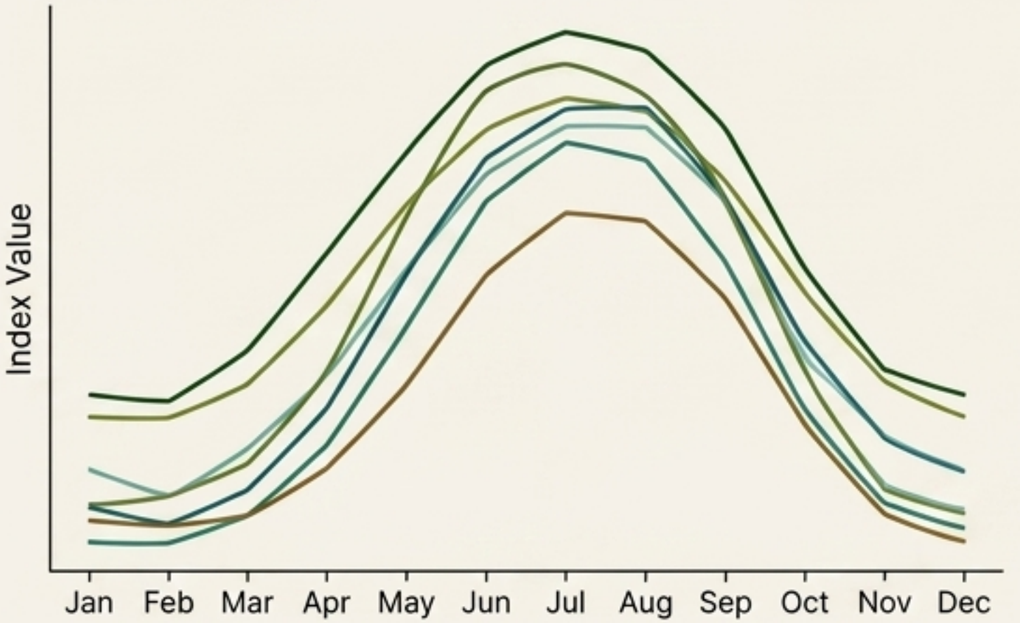
Step 1: Dual Metric Extraction for Every Point

AlphaEarth Foundations



A 64D **vector** (**A_00** to **A_63**) summarizing a full year of multi-sensor satellite observation. It captures **structural** and **spatial** context. A linear algebra representation of “**place**”.

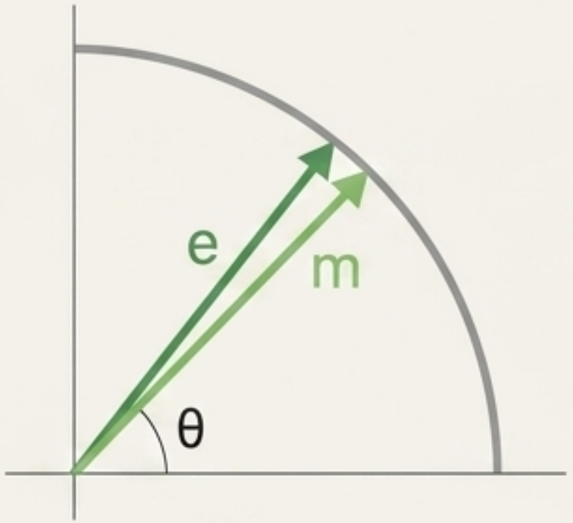
Sentinel-2 Time Series



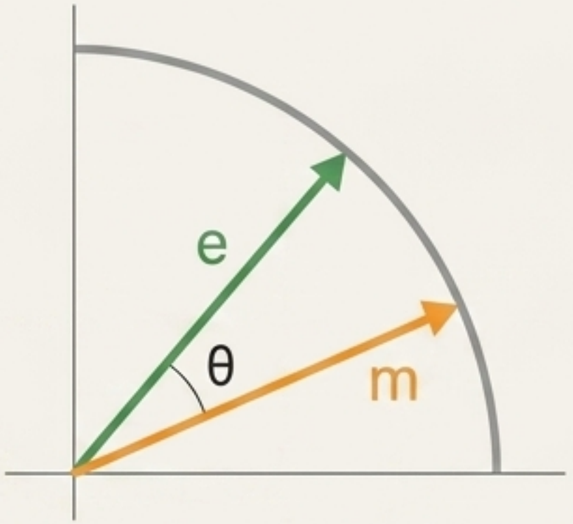
Monthly **compositions** (2020-2025) of 6 spectral **indices** (NDVI, EVI, NDII, NDWI, PSRI, CAI) at 10m resolution. Captures seasonal dynamics like **vigor**, **moisture**, **and senescence**.

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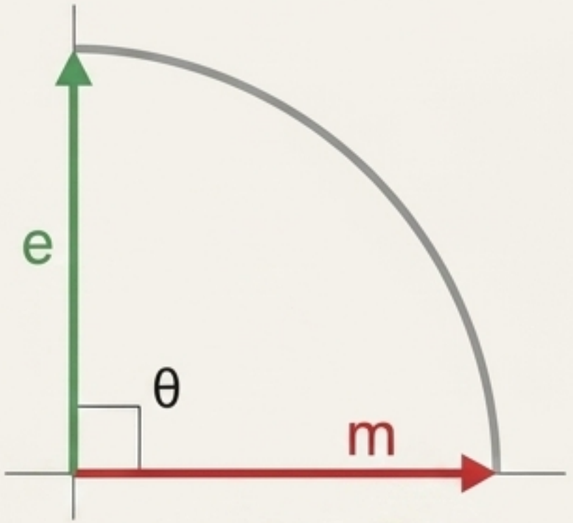
Step 2, Phase 1: Measuring Similarity (The Dot Product)



High Similarity
(cos θ ≈ 0.95)



Moderate Similarity
(cos θ ≈ 0.57)



Orthogonal / No Relation
(cos θ = 0)

$$\cos\theta = \frac{E \bullet M}{\|E\|_2 \times \|M\|_2}$$

How do we know if a candidate point acts like a field point? By calculating the Dot Product (Cosine Similarity) of their 64D AlphaEarth embeddings, we determine structural behavior. The Top-1 match defines the preliminary assigned class.

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The ‘md3’ Confidence Metric (Nearest Neighbors)

We look beyond the Top-1 match. We select the 3 Embrapa field samples most similar to our candidate to establish consensus.

3

High Confidence (md3 = 3)

All 3 of the nearest Embrapa matches share the exact same pasture typology. Highly consistent spatial neighborhood.

2

Moderate Confidence (md3 = 2)

2 out of the 3 nearest matches agree on the pasture class.

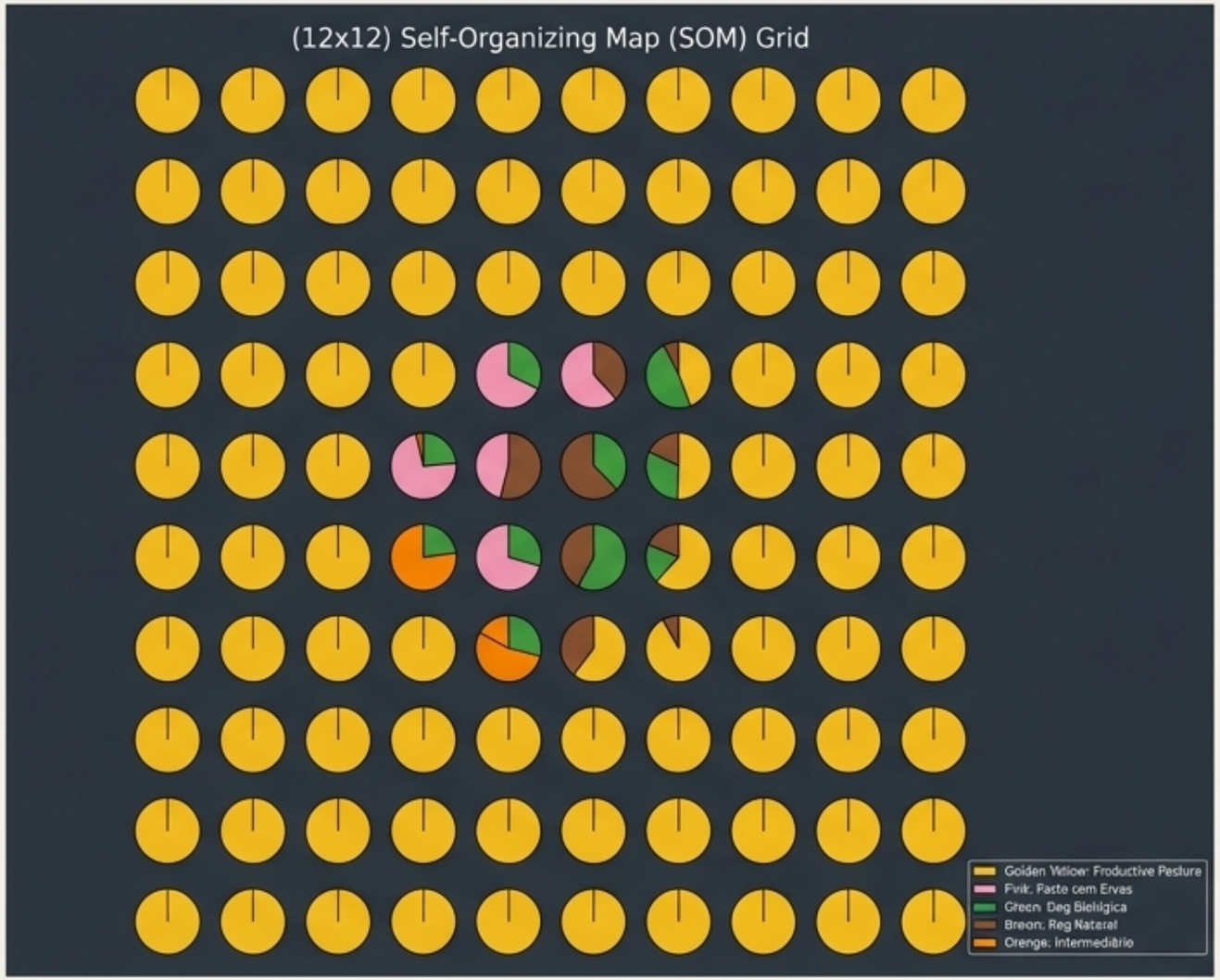
1

Low Confidence (md3 = 1)

Only the Top-1 match assigns the class; the other two diverge. Indicates an ambiguous spectral profile.

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Step 3, Phase 2: Self-Organizing Maps (Temporal Clustering)



The Mechanism

We project the 6-index Sentinel-2 time series into a 144-neuron (12x12) grid.

The Function

Points with similar temporal trajectories (phenology) cluster together, regardless of their spatial distance.

The Check

If a candidate point was labeled “Productive Pasture” by the Dot Product, but lands in a SOM neuron dominated by “Degraded” field samples, we have caught a temporal anomaly.

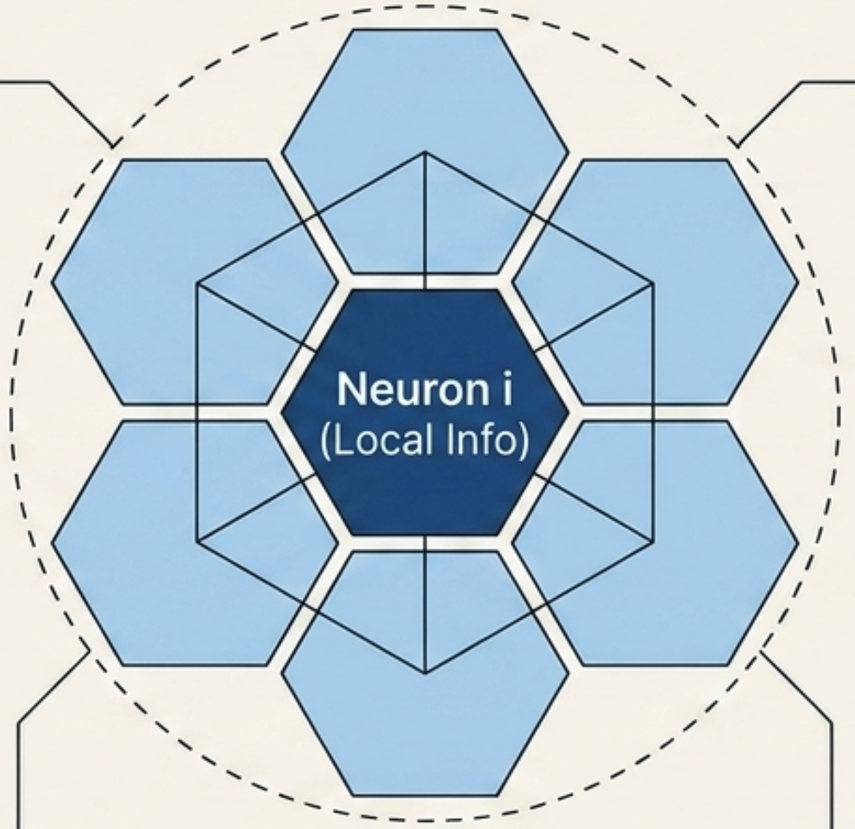
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Dirichlet-Multinomial Bayesian Smoothing

The Problem:
A single neuron might contain sparse or noisy data.

The Solution:
We calculate the **Posterior Probability** as a weighted **average** between the local neuron's count and the prior distribution of its six neighbors.

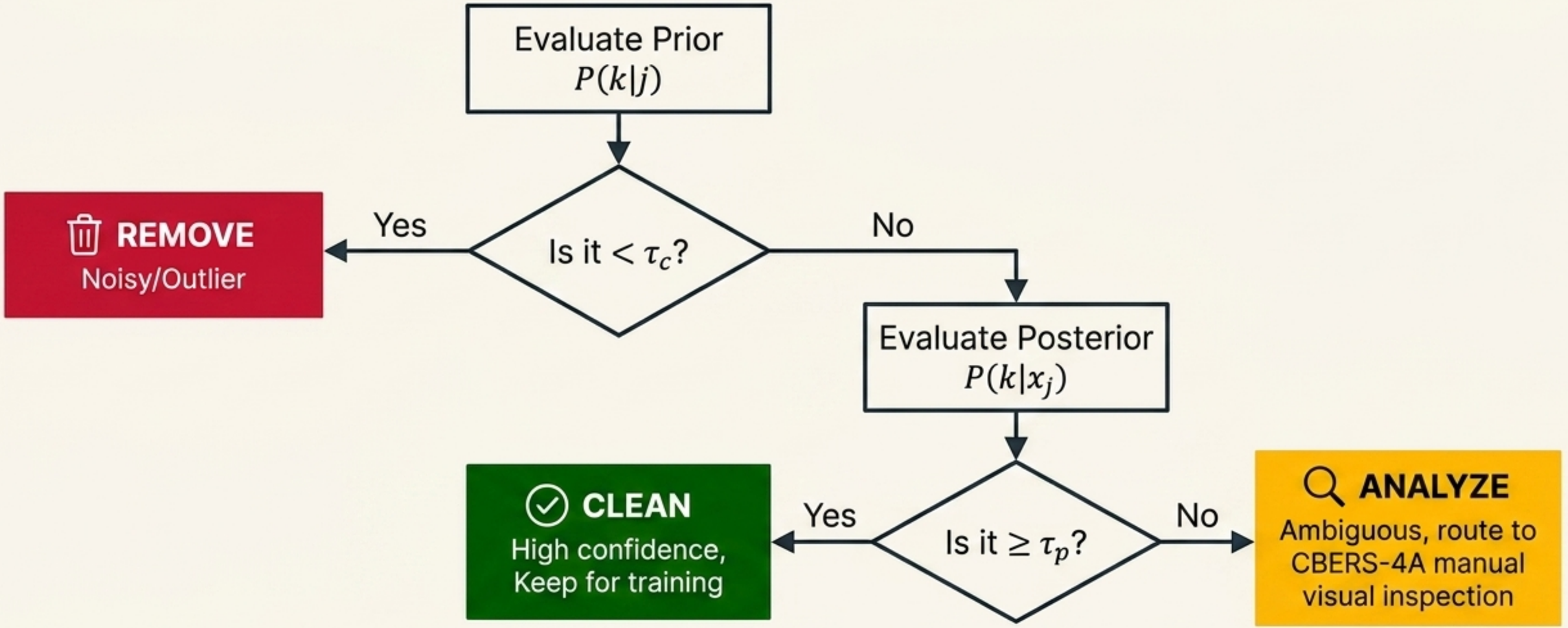
Hexagonal Neighborhood
(Spatial Smoothing)



The Result:
Anomalous, isolated data points are ‘smoothed’ out by the consensus of their surrounding temporal neighborhood, preventing mislabeled samples from skewing the class.

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The Decision Matrix: Translating Probabilities to Action



Key Takeaway: Every candidate point must pass a topological threshold.
We do not force a classification; we actively reject ambiguity.

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Diagnostic Matrix: Why Both Filters Are Required

Phase 1: Embedding Filter	Metric: 64D AlphaEarth Vectors. Mechanism: Dot Product / 1-Nearest Neighbor.	“The Broad Net” — Finds spatial/structural twins to the field samples. Ensures representation across the biome. Blind Spot: Cannot detect if the area’s use changed between 2020 and 2025.
Phase 2: SOM Filter	Metric: S2 Time Series (NDVI, EVI, etc.). Mechanism: Bayesian Hexagonal Neighborhoods.	“The Fine Sieve” — Validates temporal/phenological coherence over a full year. Resolves spectral ambiguity. Strength: Weeds out false positives and temporal anomalies.

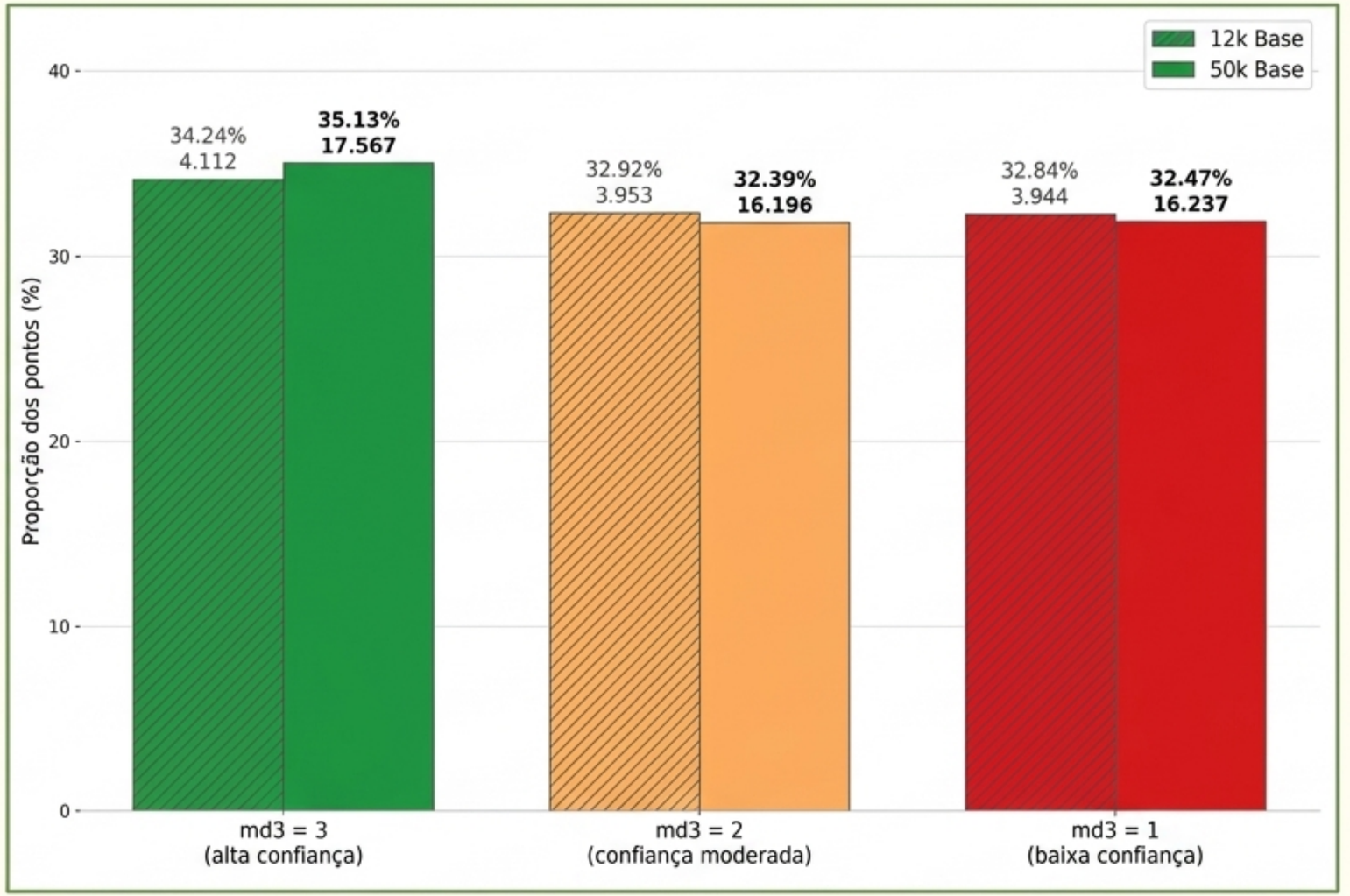
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The Output: 400% Scale Without Statistical Displacement

Pairs Calculated:
8.4 Million → 35.05 Million
(4x Scale)

Average Dot Product:
0.8217 → 0.8245
(Stability)

High/Mod Confidence (md3 ≥ 2):
67.16% → 67.52%
(Consistency)

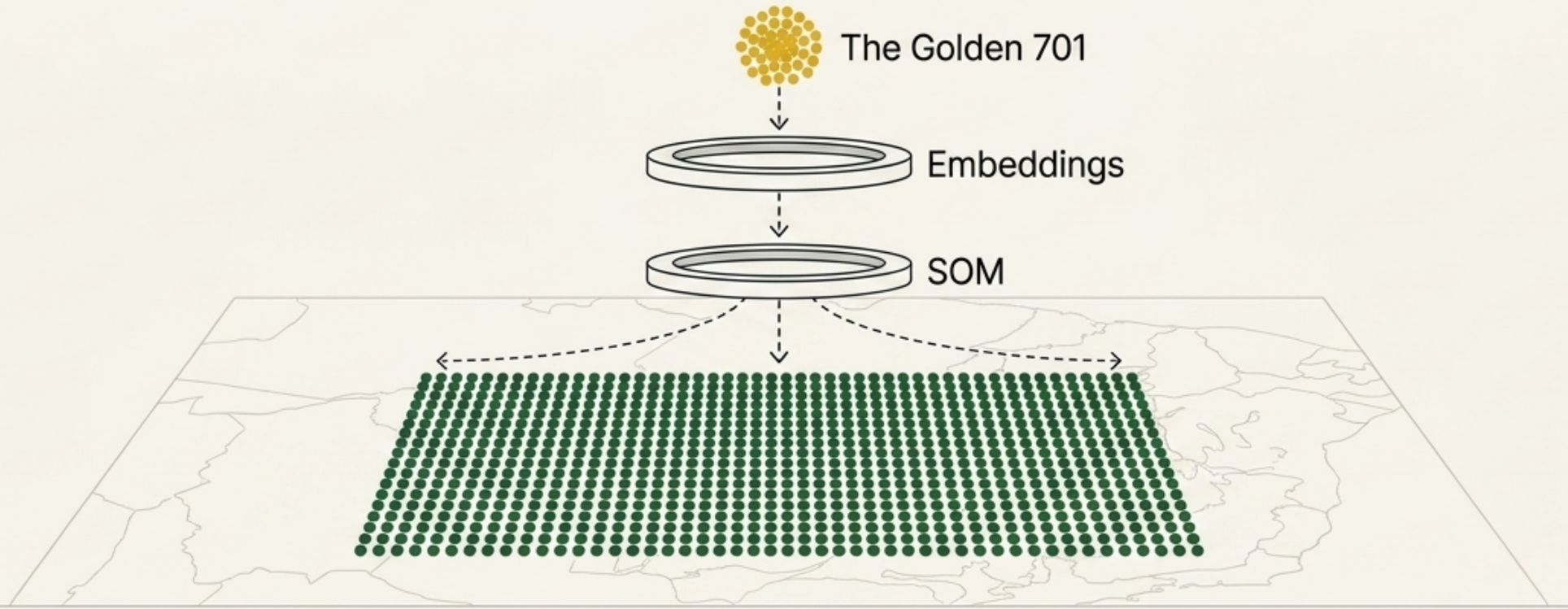


The method proves mathematically robust. By expanding to 50,000 MapBiomas points against 701 Embrapa samples, we quadrupled the dataset. Crucially, the global similarity distribution and confidence profiles remained practically identical, proving we scaled volume without introducing bias.

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Final Synthesis: Ready for Scale



Grounded in Reality

Anchored entirely by 701 expert-verified field samples.

Mathematically Vetted

Every propagated point passes rigorous spatial similarity and temporal smoothing checks.

Ready for Scale

Outputs a continuous spectrum of vetted data perfectly formatted to train the 10m Random Forest classifier.

Next Step: 10m High-Resolution Pasture Mapping and Degradation Modeling.

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